METRICMIND

Semantic Business Intelligence Engine

PROJECT REPORT

A data analytics application for transforming sales records into meaningful business metrics through a Python backend, semantic metric layer, and interactive web dashboard.

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GitHub Repository: https://github.com/MasimukkuSravani14/MetricMind

Executive Summary

MetricMind is a prototype Business Intelligence application that converts a sales CSV dataset into structured business metrics and presents them through a web dashboard. The project demonstrates a practical analytics workflow: data preparation with Python and Pandas, storage in SQLite, reusable metric definitions in a semantic layer, API access through FastAPI, and a Next.js frontend.

The implemented version focuses on deterministic sales analytics. It provides total revenue, total profit, overall profit margin, and grouped revenue views by region, country, and product. A live dashboard route consumes the backend API. Natural-language querying and LLM-based agent behaviour are identified as future enhancements rather than completed capabilities.

1. Introduction

Organizations collect sales data across products, regions, and countries. A raw spreadsheet or CSV file is useful for storage, but business users need summarized indicators to understand performance. MetricMind addresses this need by creating a small, modular BI workflow that prepares data, calculates consistent metrics, exposes them through APIs, and visualizes the results.

2. Problem Statement

Manual analysis of sales records can be repetitive and inconsistent. Different reports may calculate profit or profit margin differently, and users may need to write queries repeatedly to compare regions or products. The project aims to centralize metric definitions and make the resulting analysis accessible through a dashboard.

3. Objectives

- Read and prepare sales data from a CSV file.
- Calculate profit and profit margin from revenue, material cost, and shipping cost.
- Persist processed records in an SQLite database.
- Define reusable metrics and dimensions in a semantic layer.
- Expose metrics through REST endpoints built with FastAPI.
- Display summary KPIs and grouped results in a Next.js dashboard.

4. Technology Stack

Technology	Role in the project
Python	Core language for data processing and backend logic.
Pandas	Loads CSV data and performs tabular transformations.
SQLite	Stores the processed sales table locally.
FastAPI + Uvicorn	Provides and runs the REST API service.
Next.js / React / TypeScript	Builds the browser-based dashboard interface.

5. System Architecture

The application follows a layered architecture. The CSV file is the input source. A Python data-loading script calculates derived fields and writes the result to SQLite. The semantic layer defines approved metrics and dimensions and executes aggregation queries. FastAPI exposes those results as HTTP endpoints. The Next.js frontend calls the API and renders dashboard cards and grouped summaries.

Sales CSV **data/sales.csv**
Python + Pandas **data preparation and derived columns**
SQLite database **database/metricmind.db**
Semantic Layer **metric definitions + grouped queries**
FastAPI REST API **/metrics and /metrics/{metric}**
Next.js Dashboard **KPI cards + grouped views**

6. Data and Metric Definitions

The sample sales dataset contains 12 records and the following source fields: date, country, region, product, revenue, material_cost, and shipping_cost. The loading process adds profit and profit_margin as derived fields.

Metric	Definition
Profit	Revenue – Material Cost – Shipping Cost
Profit Margin	(Total Profit ÷ Total Revenue) × 100
Revenue	Sum of revenue values

Available dimensions: region, country, and product. The semantic layer validates requested metric and dimension names before executing the corresponding SQL aggregation.

7. API Design

Method and endpoint	Purpose
GET /	Returns API status information.
GET /metrics	Lists supported metrics and dimensions.
GET /metrics/{metric}?group_by={dimension}	Returns a metric overall or grouped by a supported dimension.

8. Implementation and Execution

Backend setup (Windows PowerShell)

```
cd C:\MetricMind
python -m venv .venv
.venv\Scripts\activate
pip install -r requirements.txt
python load_database.py
python -m uvicorn api:app --reload
```

Open the API documentation at http://127.0.0.1:8000/docs. Keep this terminal running while using the frontend.

Frontend setup (a second terminal)

```
cd C:\MetricMind\frontend
npm install
npm run dev
```

Open http://localhost:3000 for the dashboard. The live dashboard route is http://localhost:3000/live.

9. Observed Results

During the local run, the dashboard successfully connected to the FastAPI backend. The observed sample outputs were:

Indicator	Observed value
Total Revenue	601,000
Total Profit	202,500
Overall Profit Margin	33.69%
Revenue by Region	Europe: 366,000; Asia: 235,000
Revenue by Product	Laptop: 253,000; Phone: 248,000; Tablet: 100,000

Note: The dataset does not specify a currency symbol, so the values are reported without assigning a currency denomination.

10. Current Scope and Limitations

- The current implementation uses a small sample sales dataset for demonstration.
- The dashboard is connected to the implemented FastAPI metrics endpoints.
- Natural-language-to-SQL or LLM-based agent functionality is not yet implemented.
- Automated anomaly detection, authentication, deployment, and production monitoring are future work.
- The project is a local prototype and has not been presented as a production-scale BI platform.

11. Future Enhancements

- Add a natural-language query interface using an LLM with controlled metric and dimension access.
- Introduce AI-generated explanations and business recommendations.
- Add anomaly detection and data-quality checks.

- Support larger datasets, additional connectors, and scheduled refresh.
- Add user authentication, deployment, tests, and monitoring.

12. Conclusion

MetricMind demonstrates an end-to-end foundation for semantic business intelligence. It connects data preparation, database storage, consistent metric definitions, API delivery, and dashboard visualization. The modular design provides a clear base for future AI-assisted analytics capabilities.

References and Project Link

Project repository: https://github.com/MasimukkuSravani14/MetricMind
FastAPI documentation: https://fastapi.tiangolo.com/
Pandas documentation: https://pandas.pydata.org/docs/
Next.js documentation: https://nextjs.org/docs